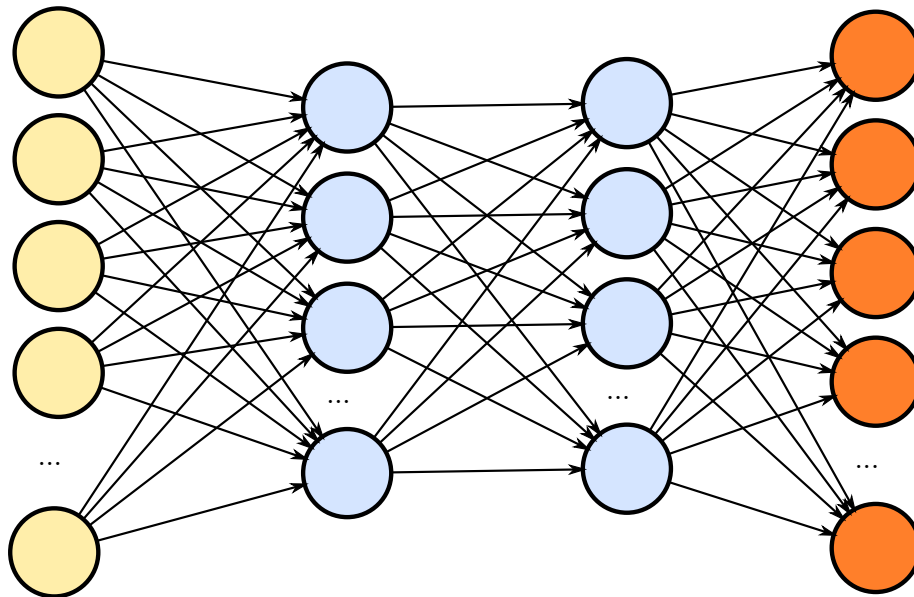

Deep Learning

TAMU CSCE-636

Brody Jordan

Fall 2025



Acknowledgements

These notes were compiled from the course materials for CSCE-636 (Deep Learning) taught by Dr. Anxiao Jiang at Texas A&M University during the Fall 2025 semester. It is worth mentioning that this course was ONLINE so lecture dates are not consistent and fixed. Each *chapter*, therefore, is based on material sectioning outline in the course syllabus. Any significant diverges from the course syllabus, whether at my own discretion or due to changes in the course structure, will be noted here and in the preamble of each chapter.

Deep Learning (CSCE-636)

 **Instructor:** Dr. Anxiao Jiang
 **Semester:** Fall 2025
 **Time:** ONLINE
 **Location:** ONLINE

Description: Basic concepts in neural computing; functional equivalence and convergence properties of neural network models; associative memory models; associative, competitive and adaptive resonance models of adaptation and learning; selective applications of neural networks to vision, speech, motor control and planning; neural network modeling environments.

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Chapter 1

Introduction

Deep learning is a subarea of machine learning, just as machine learning is a subarea of artificial intelligence. In artificial intelligence, we want to learn all kinds of topics about how to create and understand intelligence using computer systems. In this broad area, there is a particular method called machine learning where we use lots of data to train a machine, so the machine may learn how to solve a problem. Deep learning is a particular kind of machine learning where we use deep neural networks for learning.

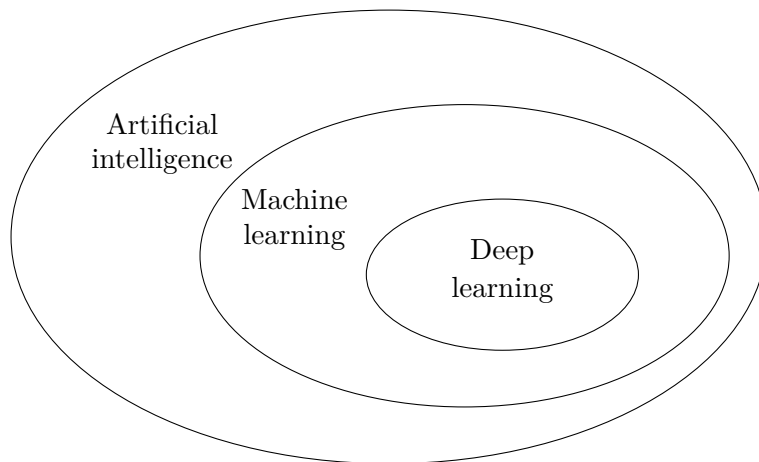


Figure 1.1: Relationship between AI, ML, and DL

Classical algorithms take rules and data to produce answers. In contrast, machine learning algorithms take data and answers to produce rules.